FINM3013 Introduction to Financial Derivatives

Chapter 04

Solution to Class exercises

1. A. The equivalent rate with semiannual compounding is $2\times(e^{0.05/2}-1) = 0.0506$ or 5.06%.

2. A. The equivalent rate per quarter is $\sqrt{1.06}-1 = 2.956\%$. The annualized rate with quarterly compounding is four times this or 11.83%.

3. D. The forward rate for the third year is $(3\times0.065-2\times0.06)/(3-2) = 0.075$ or 7.5%.

4. A. The forward rate is the interest rate implied by the current term structure for future periods of time. For example, earning the zero rate for one year and the forward rate for the period between one and two years gives the same result as earning the zero rate for two years.

5. A. The six month rate is 1.5% per six months. The one year rate is 2% per six months. The one year par yield is the coupon that leads to a bond being worth par. A is the correct answer because $(3.99/2)/1.015+(100+3.99/2)/1.02^2 = 100$. The formula in the text can also be used to give the par yield as $[(100-100/1.02^2)\times2]/(1/1.015+1/1.02^2)=3.99$.